ITA14 - ETHICAL HACKING
EX - 8

Execute and display the results for the following Nmap service version and OS detection commands in Kali Linux operating systems:
i) detect the version of services running
ii) Aggressive scan
iii) Detect operating system of the target

Aim
To execute and display the results for the following Nmap service version and OS detection commands in Kali Linux operating systems: i) detect the version of services running ii) aggressive scan iii) detect operating system of the target

Procedure
• Open (Cloud Shell Editor)

• i) detect the version of services running

1. sudo apt update -- press Enter
2. sudo apt install nmap -y -- press Enter
3. nmap --version -- press Enter
4. nmap -sV scanme.nmap.org -- press Enter

• ii) aggressive scan

1. nmap -A scanme.nmap.org -- press Enter

• iii) detect operating system of the target

1. sudo nmap -O scanme.nmap.org -- press Ent

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┌──(Message from Kali developers)

 This is a minimal installation of Kali Linux, you likely
 want to install supplementary tools. Learn how:
 ⇒ https://www.kali.org/docs/troubleshooting/common-minimum-setup/

┌──(Run: “touch ~/.hushlogin” to hide this message)
┌──(kali㉿Harshini)-[~]
└─$ sudo apt update
[sudo] password for kali:
Hit:1 http://http.kali.org/kali kali-last-snapshot InRelease
161 packages can be upgraded. Run 'apt list --upgradable' to see them.

┌──(kali㉿Harshini)-[~]
└─$ sudo apt install nmap -y
nmap is already the newest version (7.99+dfsg-1kali1).
Summary:
  Upgrading: 0, Installing: 0, Removing: 0, Not Upgrading: 161

┌──(kali㉿Harshini)-[~]
└─$ nmap --version
Nmap version 7.99 ( https://nmap.org )
Platform: x86_64-pc-linux-gnu
Compiled with: liblua-5.4.8 openssl-3.6.2 libssh2-1.11.1 libz-1.3.1 libpcre2-10.46 libpcap-1.10.6 nmap-libdnet-1.18.0 ip
v6
Compiled without:
Available nsock engines: epoll poll select

┌──(kali㉿Harshini)-[~]
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9929/tcp open     nping-echo   Nping echo
Service Info: OS: Linux; CPE: cpe:/o:linux:linux_kernel

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 50.05 seconds

┌──(kali㉿Harshini)-[~]
└─$ sudo nmap -O scanme.nmap.org
Starting Nmap 7.99 ( https://nmap.org ) at 2026-11-08 16:17 +0530
Nmap scan report for scanme.nmap.org (45.33.32.156)
Host is up (0.25s latency).
Other addresses for scanme.nmap.org (not scanned): 2600:3c01::f03c:91ff:fe18:bb2f
Not shown: 993 closed tcp ports (reset)
PORT     STATE    SERVICE
22/tcp   open     ssh
25/tcp   filtered smtp
80/tcp   open     http
135/tcp  filtered msrpc
139/tcp  filtered netbios-ssn
445/tcp  filtered microsoft-ds
9929/tcp open     nping-echo
OS fingerprint not ideal because: Host distance (24 network hops) is greater than five
No OS matches for host
Network Distance: 24 hops

OS detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 54.01 seconds
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```
Compiled without:
Available nsock engines: epoll poll select

┌──(kali㉿Harshini)-[~]
└─$ nmap -sV scanme.nmap.org
Starting Nmap 7.99 ( https://nmap.org ) at 2026-11-08 16:16 +0530
Stats: 0:00:13 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan
SYN Stealth Scan Timing: About 11.60% done; ETC: 16:17 (0:01:31 remaining)
Stats: 0:00:16 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan
SYN Stealth Scan Timing: About 15.38% done; ETC: 16:17 (0:01:23 remaining)
Stats: 0:00:45 elapsed; 0 hosts completed (1 up), 1 undergoing Script Scan
NSE Timing: About 97.10% done; ETC: 16:16 (0:00:00 remaining)
Stats: 0:00:48 elapsed; 0 hosts completed (1 up), 1 undergoing Script Scan
NSE Timing: About 83.33% done; ETC: 16:16 (0:00:00 remaining)
Nmap scan report for scanme.nmap.org (45.33.32.156)
Host is up (1.5s latency).
Other addresses for scanme.nmap.org (not scanned): 2600:3c01::f03c:91ff:fe18:bb2f
Not shown: 993 closed tcp ports (reset)
PORT     STATE    SERVICE      VERSION
22/tcp   open     ssh          OpenSSH 6.6.1p1 Ubuntu 2ubuntu2.13 (Ubuntu Linux; protocol 2.0)
25/tcp   filtered smtp
80/tcp   open     http         Apache httpd 2.4.7 ((Ubuntu))
135/tcp  filtered msrpc
139/tcp  filtered netbios-ssn
445/tcp  filtered microsoft-ds
9929/tcp open     nping-echo   Nping echo
Service Info: OS: Linux; CPE: cpe:/o:linux:linux_kernel

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 50.05 seconds
```